FAST BREEDER REACTOR PROGRAMME IN THE UNITED STATES

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1. Introduction

The primary objective of the civilian power reactor development programme in the United States is to achieve widespread commercial use of nuclear energy for the production of heat and electricity while fully exploiting the energy available in our resources of uranium and thorium. The Commission's objective also includes fostering the development of a self-sufficient and competitive nuclear industry.

The successful achievement of the primary objective will require the development of fast breeder reactors and their successful introduction into commercial use. To accomplish this, one of the highest priorities has been given by the Commission to planning and executing the research and development programme required for mastering the technically difficult and challenging sodium cooled fast breeder reactor concept in a timely manner. This priority was determined after an assessment was made of the status of various breeder concepts and of the overall liquid metal fast breeder reactor (LMFBR) programme and each of its programme elements. On a lesser priority basis, work is continuing on other coolant concepts.

Based on this assessment, formal intensive and comprehensive reviews are being made in each technical area of the LMFBR programme to determine the major needs of the programme, its deficiencies, and the plans required to eliminate these deficiencies and further strengthen the programme. The initial reviews have been completed on fuels and materials, physics, and their associated facilities. Reviews are continuing in the areas of fuels reprocessing and recycling, safety, components, instrumentation and control, sodium technology, fuel handling, plant design, and in each case, the related facilities. Though development work is continuing in each of the other areas, and has in part been substantially augmented, the results of each formal review are

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required to bring into proper perspective the parts requiring strengthening and/or augmentation, including interactions with other programme elements which may be affected by the review. We are convinced that each of these commitments involving large resources of personnel, funds and facilities, and many organizations cannot be accomplished successfully and in a timely manner unless it is planned in a deliberate and disciplined way with complete attention to details and engineering practices. Consequently, considerable emphasis is being placed on assessing the specific status of the programme, defining its needs, and aggressively initiating those actions required to strengthen the fast breeder reactor programme, including the related engineering practices. This assessment has covered matters such as adequacy of: (1) management of programme planning and technical direction; (2) resources to accomplish the programme; (3) participation of the laboratories, industry, and the utilities; (4) criteria and standards; (5) development of industrial capability; and (6) attention to detail design, fabrication, test, and operational features of hard commitments.

The detailed analyses and evaluations so far completed of the two LMFBR programme elements, namely the fuels and materials programmes, the physics programme, and their related facilities, resulted in conclusions that each of these elements required significant strengthening and augmentation. Implementation of the resulting recommendations is being initiated with the advice and co-operation of industry. In addition, the AEC has established an LMFBR programme office at ANL which is serving as an extension of the Division of Reactor Development and Technology to assist in LMFBR programme evaluation and planning. In this way, formal plans for the other elements of the programme will be developed with continued consultation with industry and the utilities.

The objective of the fast breeder reactor effort is to develop a safe, reliable, and economic power plant for the utility environment. But a statement of the objective indicates only the problem, not its nature or its magnitude. For example, the fast breeder reactor requires a highly subdivided and enriched fuel. The concentration of a critical mass in a small volume requires heat removal capabilities which can be furnished by sodium. Sodium also has excellent neutronic characteristics. However, the large fast flux, the characteristics of sodium, and the resultant high temperatures and temperature differentials result in a complex reactor whose difficulty of successful, timely, and economic accomplishment should not be underestimated.

A review of the status of the nuclear industry in the United States may help place the fast breeder development programme in better perspective. Any such review should include a projection of nuclear growth and a comparison of its status with predictions, trends in nuclear reactor size, trends in capital costs, and estimates of the AEC civilian power reactor research and development expenditures.

A most significant trend has been evident from the recent orders by the electric utility companies in the United States for nuclear power plants during 1965 and the first 4 months of 1966 (Fig. 1). During this period, 25% of the new steam electric plant generating capacity orders was nuclear, with about 4700 MW(e) new commercial nuclear plants ordered in 1965 exceeding by a factor of five the total capacity of all commercial nuclear plants ordered prior

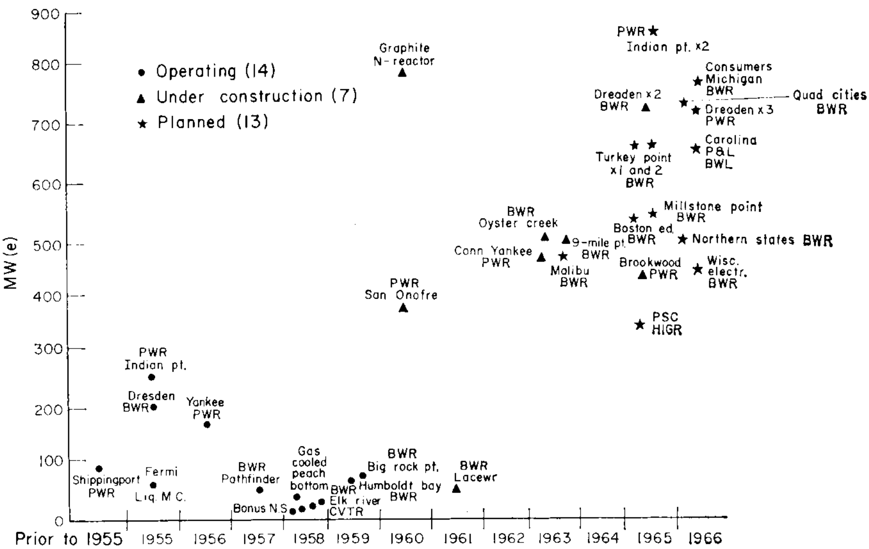

FIG. 1. Nuclear power plants, trends in reactor size (based on orders placed April 1966).

to 1965 in the United States. The procurement of six nuclear plants totalling about 3700 MW(e) has already been announced in 1966, with these nuclear plants representing one-half of all the new steam generating capacity orders thus far this year.

The recent orders for nuclear plants by the electric utility companies have significantly exceeded the projections of nuclear growth, contained in the Report to the President in 1962 on the civilian nuclear power programme. Utility plans now show that by 1973 there will be about 36,000 MW(e) net plant capacity as compared with about 10,000 MW(e) net estimated by the AEC in the 1962 Report to the President.

The 1962 Report is recognized as an important reference document, prepared after one of the most publicized formal reviews by the United States Atomic Energy Commission of the nuclear power requirements and the reactor development efforts. It requires continual review and updating, and such a review is now in progress.

One obvious result of the increasing number of orders for nuclear plants is that the increased uranium consumption and plutonium production will have an immediate impact on the conservation of our nuclear resources. This, too, re-emphasizes the increasing importance of the fast breeder programme.

Another interesting aspect of the recent increased procurement of nuclear power plants is the pronounced trend towards larger size reactor units, as shown in Fig. 1, increasing from a range of 50–100 MW(e) to 400–800 MW(e)

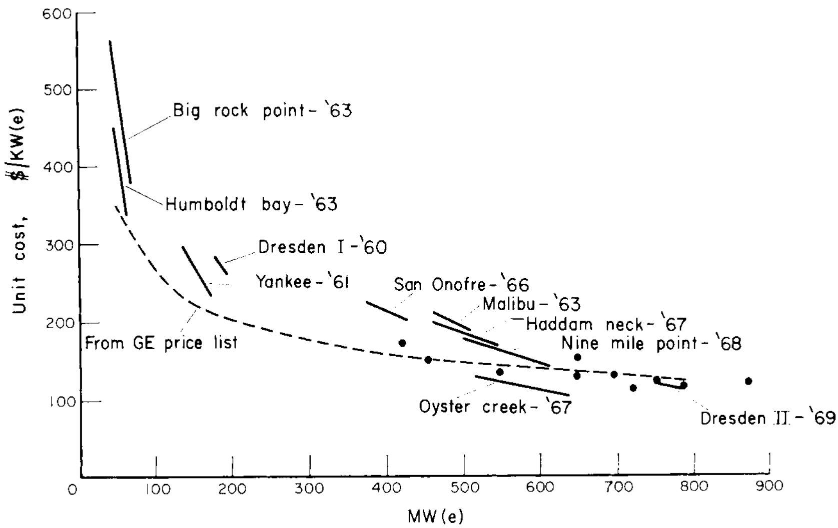

FIG. 2. Trends in capital costs. Light water nuclear plants.

over the 10-year period. It is quite possible that the present 1000 MW(e) plant, which is the size towards which our present R&D effort is directed, may soon be reached, and we may have to accelerate our investigation of larger target plants, such as 1500 and 2000 MW(e) sizes.

The downward trends in capital cost for commercial light water reactor plants are shown in Fig. 2. The pronounced reduction in cost from over $400 to about $100–$150 per kW(e) installed is owing to advances in technology, increases in unit size, and increasing volume of business. A similar cost trend should be expected for fast breeder plants as the breeder reactor programme develops and these plants are accepted by the utilities.

Recent strengthening and augmentation of the fast breeder programme and consideration of other priorities has resulted in an adjustment in the Government's funding for the civilian power programme, with a significant increase in the LMFBR programme. Table 1 indicates that the budget for the fiscal year 1965–6–7 period for the fast breeder operating costs alone, excluding facilities, has increased 90% within 2 years. In addition, important facility expenditures, authorized or requested during the fiscal year 1965–7 period, include those for the Zero Power Plutonium Reactor critical assembly, the Fast Flux Test Facility, the Sodium Pump Test Facility, and accelerators required to measure nuclear cross-sections, such as the fast neutron generators at the Argonne and Oak Ridge National Laboratories.

TABLE 1. FAST BREEDER REACTORS
Summary—operating costs

	(In millions)		
	Actual FY 1965	Estimate FY 1966	Estimate FY 1967
Liquid metal cooled reactors	$	$	$
Fast breeder fuel development	5·3	7·3	11·3
Fast breeder reactor physics	5·2	5·6	9·8
Liquid metal component development	4·4	5·4	7·0
Design concept anal. and adv. system evaluation	0·5	1·4	1·1
Experimental Breeder Reactor-II (EBR-II)	5·6	6·6	6·4
Fast fuel test facilities	2·5	7·0	8·9
TOTAL:	23·5	33·3	44·5
Other fast breeder reactor systems			
Fuel development	0·3	0·6	0·4
Reactor physics	*	-0-	0·2
Component development	-0-	-0-	0·1
Design concept anal. and system evaluation	0·2	0·1	0·5
TOTAL:	0·5	0·7	1·2
TOTALS:	$24·0	$34·0	$45·7

* Actual = $10,000.

As each review now being made in the fast breeder reactor programme ends and results from the programme become available, there will undoubtedly be need for various adjustments, and technical planning and expenditure plans will be revised accordingly.

2. The Fast Breeder Programme in the United States

Although the need for fast breeder reactors has been reaffirmed for many years, a summary is in order.

The fast breeder can provide the most efficient means of exploiting the energy available in uranium. It minimizes the quantity of uranium consumed per unit of electricity generated and, at the same time, provides the potential for low fuel costs. It permits 60–90% utilization of the uranium processed from the ore, in effect extending the ore reserves manyfold, while providing for potential fuel costs of less than 1 mill/kWh of electricity because of fuel burnup capabilities.

The choice of plutonium as the fuel for the fast breeder is based on its high rate of neutron production, its relatively advanced state of technology as compared to thorium, and the availability of uranium, at reasonable prices, to fuel thermal reactors which, in turn, will provide the plutonium for the first cores of the fast breeder system.

It is appropriate that a discussion of the LMFBR programme proceed on the same basis as the programme planning, by elements of the programme. These major elements include fuels and materials, fuel reprocessing and recycling, physics, safety, components, instruments and control, coolant technology, fuel handling, plant design including systems, and related facilities.

3. Fuels and Materials

The fast breeder reactor has the capability of utilizing enriched plutonium–uranium fuels to burnups of over 100,000 MW days per metric ton of heavy metal for ceramics and over 50,000 MWd/t for alloys. The operation of 15–30 kW/ft of fuel pin in sodium temperature of 650°C and higher requires a small diameter pin with central temperatures held below fuel melting. Thus, the objectives of the LMFBR fuels and materials programme are a reliable, safe, high performance, and ultimately a high burnup fuel, with a target of a 3-year average life. There must be no meltdown, slumping, ratcheting, clad rupture, or fuel redistribution, and no communication to other fuel in the event of failure. Cladding temperatures during normal operation may reach 760°C with an exposure over 10^{23} nvt. Ceramic fuel temperatures will be over 2200°C with 650°C average sodium outlet temperature.

The safety, reliability, and economic requirements meant that the fuels and materials programme, as a major LMFBR element, had to be extensively reviewed, and such a review was made during 1965. It revealed a significant lack of meaningful data from in-pile performance. This paucity of positive technical information meant there were little sound criteria by which to judge experimental data on fuels and materials properties and by which the performance of fuels and cladding materials in the LMFBR environment could be adequately predicted. Thus, a need was noted for the identification and establishment of fuels and materials criteria and standards. Based on the review, a strengthened and augmented R&D programme was formulated which should successfully meet its requirements.

It was concluded from the review that ceramic fuels, though lacking significant fast flux and sodium environment data, have the best potential to reach LMFBR objectives. The two prime ceramic fuel candidates are plutonium–uranium oxides and carbides. Alloy work is continuing on a reduced scale with fuel development criteria being determined upon which to base future evaluations. Research and development to a lesser extent has been initiated on plutonium–uranium nitrides, sulphides, phosphides, and combinations. An evaluation has been under way to determine future efforts on the tantalum encapsulated liquid plutonium fuel.

The fuels and materials review has resulted in increased emphasis on development of fuel cladding materials since reliable containment of fuels in an adequate cladding material in the LMFBR environment is a major problem.

The plutonium–uranium oxides are currently favoured by industry in the United States and major foreign programmes. With these fuels, irradiations in the United States have been performed to approximately 9 at.% burnup in a thermal flux, and fast flux irradiations in EBR-II have reached 1·5 at.% and are continuing.

The plutonium–uranium carbides, when compared to the oxides, offer a potential improvement in breeding through their higher metal atom density and higher fuel performance (as a result of the higher thermal conductivity). Irradiations of the carbides in the United States have exceeded 10 at.% in a thermal flux. Fuel swelling, fuel-to-clad compatibility and control of fuel composition are the major problems to be investigated.

The alloy fuels have a higher metal density, thermal conductivity, thermal coefficient of expansion, and breeding gain potential when compared with the ceramics. Fuel-to-clad compatibility, fuel swelling, and low melting temperatures have been the major problems encountered with uranium–plutonium–fissium alloys. Recent investigations with the U–Pu–Zr ternary alloys have indicated improvement in the three aforementioned problem areas. The fuel is compatible with AISI 304 SS up to 800°C, and burnups in excess of 5 at.% have been achieved in a thermal flux.

The programme on fuels and cladding materials in the United States will be paced to a significant degree by the existing fast flux irradiation facilities. The fast flux testing space in the United States' reactors, EBR-II and Fermi, is limited, and thermal reactors are used extensively for screening and comparative purposes. From a fuel cycle economic standpoint, burnups of ceramic fuels to over 10 at.% (of heavy metal atoms) are required. At these burnups, the cladding materials will receive neutron doses of greater than 10^{23} nvt (neutrons per cm^2). In the EBR-II core, the average neutron flux is about $2{\cdot}0 \times 10^{15}$ nv (neutrons per cm^2/s). Therefore, to achieve burnups and cladding doses on candidate materials typical of large sodium cooled breeder plants (10 at.% and 20^{23} nvt), the residence time in EBR-II (based on a 50% plant factor and 42·5 MW(th) operation) is over 3 years for enriched fuel.

The programme is currently emphasizing fuel synthesis and fabrication methods, engineering properties, fuel-clad-coolant compatibility, performance limits, and fast flux irradiations. These in-pile tests, together with the attendant out-of-pile programme, will establish those parameters which significantly affect fuel performance. The programme is planned to provide data to include: (1) the optimum fuel density and best porosity arrangement; (2) irradiated fuel swelling and fission gas release rates; (3) plutonium migration and concentration; (4) the kW/ft rating which results in fuel movement; (5) the effects of transients on fuel and clad; (6) the effect on the clad of a fast neutron and sodium coolant environment; (7) the consequences of defective fuel operation; and (8) the basic mechanical, physical, and chemical properties of unirradiated and irradiated fuel.

The fuel development programme for the next few years will concentrate on the plutonium–uranium oxides, carbides and alloys to determine which fuel systems will be emphasized in the following years. It is recognized that the most promising fuel type available at the time will be selected for the first demonstration plant and that a concentrated development programme will have to be undertaken to verify the performance of that fuel by a test programme utilizing a statistically significant number of fuel sub-assemblies. The development efforts on the other candidate fuels will be maintained in addition to supporting applied research and exploratory development for the long range.

Satisfactory performance of candidate fuels may be realized if cladding integrity can be assured. The cladding materials being considered include austenitic stainless steels, nickel base alloys, and refractory alloys.

More pertinent information is available for the stainless steel alloys than for any of the other materials presently being considered as potential cladding candidates. Continued emphasis on stainless steels in the clad materials development is justified by the performance observed so far in high temperature sodium, by the apparently good compatibility with plutonium–uranium oxide fuel, and by its relatively low cost and favourable neutronics as compared to other candidate materials. Considerably more data are needed for the stainless steels, particularly with respect to compatibility with other ceramic fuels and long-term mechanical properties in the LMFBR fast neutron environment. Approximately 60% of the cladding programme is directed towards developing the stainless steels.

The successful use of stainless steel as an LMFBR fuel cladding material is not assured at this point. There is a continuing need to investigate modifications of stainless steels and other potential cladding materials, such as the higher strength nickel alloys and refractory alloys, and there is a general need for more information on their compatibility with LMFBR fuel materials, compatibility with sodium, and long-term mechanical properties, both out-of-pile and in-pile.

There is a dearth of data on large, fast neutron dose effects on materials. Several competing neutron damage mechanisms may have deleterious effects on the iron and nickel base alloys, and to a lesser extent on other alloys. The resolution of these questions will have the highest priority among the development activities, recognizing that existing fast reactors such as EBR-II and Fermi require 3 years or more to accumulate the needed fast neutron doses.

Recognizing the limitations of existing facilities, and recognizing that demonstration plant cores can be changed, it is believed that sufficient development to select and verify the fuel and cladding for the first demonstration plant can be performed in existing irradiation facilities, assuming reasonable on-stream performance. Fuel performance and economic optimization, accelerated testing, instrumented testing and fuel element design improvements and certain safety assessments require environments similar to the normal and abnormal conditions expected in an LMFBR plant. In general, such fuel and cladding materials testing will have to await the availability of the Fast Flux Test Facility (FFTF), described in more detail below.

The fuels and materials programme is under continual review. As the other programme reviews are completed, and R&D progresses, their results will be factored into the fuels and materials programme, and, in turn, the results from the fuels and materials development programmes will be used to modify other programmes as deemed necessary. The results of the first phases of the fuels and materials programme will be statistically proven initially in the FFTF and on a larger scale in the programmes for the first demonstration plants.

4. Reprocessing and Recycling

A thorough review of the reprocessing and recycling programme is under way to determine the priorities in this programme and define those areas which require strengthening. It is recognized that potentially serious problems exist in certain segments of fast reactor fuel cycles which must be resolved before a reliable and economic system can be achieved. In this context the fuel cycle includes not only chemical reprocessing of irradiated fuel, but also conversion and refabrication of the recovered fuel for further irradiation.

In general, the fast reactor of the future will be operated to fuel burnups over fivefold greater than present thermal reactors. The fuel enrichment of fast reactors will be in the range of 10–20% plutonium containing substantial quantities of Pu-238, Pu-240, and Pu-241 which will introduce a new and important factor in the fuel cycle. The resulting high thermal performance results in a small diameter fuel rod relative to thermal reactor fuel. The enrichment, geometry, and burnup affect the fuel cycle significantly.

The current reprocessing technology that may be applied to fast reactor fuels falls into three categories, namely, the conventional aqueous process with development of head-end treatments specific to the fast reactor fuel, the volatility process now being developed may show improved economics, and the pyrochemical processes.

While much of the aqueous processing developed for thermal reactor fuels may find application to fast breeder reactor fuels, particularly if a long decay time is acceptable for the initial fuel discharged from the first demonstration plants, higher burnup in the fast system will introduce many new problems due both to the chemical effects of large quantities of fission products in the irradiated fuel and to the associated high radiation levels.

Techniques for the removal of cladding and bonding from sodium bonded fuels are being developed which should be safe and compatible with aqueous processing operations. This compatibility is being determined in terms of stability of fuel solutions, chemical effects of the large quantities of fission products on solvent extraction efficiency and product recovery, and radiation effects on organic solvents and ion exchange media.

The fluoride volatility process currently being developed may be a possible economical means for processing fast reactor fuels, particularly the ceramic type. Potential advantages of volatility over aqueous processing methods include comparatively few process operations, comparatively high fission product decontamination in relatively simple and compact equipment, the use of inorganic chemical reagents with low susceptibilities to deleterious radiation effects, virtual absence of neutron moderating chemicals in the process, and uranium and plutonium in forms amenable to fuel recycle. The potential commercial application of this technology for thermal reactor fuel requires much R&D work in the next few years. The results of this effort will help establish its potential for the fast reactor fuel cycle, particularly in view of the high level of commercial interest in this process.

Pyrochemical processes currently under development for processing metallic fast reactor fuels show promise for processing ceramic fuels. These processes involve partial decontamination of fission products followed by remote refabrication of the fuel for recycle. The interest in pyrochemical processing technology for fast fuels is based on the possibility for rapid and economical recycling of fuels after short decay storage time. In addition, such processes handle the fuels in a compact form (either as solids or as solutions in molten metals or fused salts), are not encumbered with severe critical mass restrictions (because of the absence of good moderating materials), and produce concentrated solid wastes. The first pyrochemical processes were developed and are being demonstrated for the recycling of EBR-II metallic fuels. Based on experience in this programme, advanced processes are being evaluated for plutonium bearing ceramic fuels (oxide and carbide) now under development.

Methods for integrating fuel reprocessing and feed material conversion with fuel refabrication are under development. Problems resulting from high neutron and hard gamma radiations from plutonium isotopes and other transuranium species associated with a fast neutron flux must be evaluated in the selection of refabrication shielding requirements. At this time, it is questionable if direct fabrication techniques can be applied to recycled plutonium fuel. The development of semi-remote technology and fully remote technology is currently under way, and such efforts may become extremely important to the successful development of economic fast reactor fuel cycles. Specific problems to be resolved include the development of appropriate technology to ensure operational and public safety from radiation hazard, evaluation of effects of high level (alpha and neutron) radiation on economics of the conversion and refabrication processes, and the development of appropriate quality control measures to ensure production of fuel elements within specification for recycle to the reactor.

Reprocessing costs and inventory charges constitute an important percentage of total fast reactor fuel cycle costs. A programme must be established and implemented to obtain reductions in the fast reactor fuel cycle costs. Fortunately, the specific process will not have to be selected for further exploitation at the time that the fuel type is chosen for the first demonstration plants. However, it is necessary to obtain meaningful information on the reference processes for ultimate demonstration of the demonstration plant fuels and to develop economic processes for the large economic LMFBR.

5. Fuel Handling

As for other major programme elements, a review is under way to determine the priorities of a fuel handling programme, and to define those areas which need modifying.

The design and development of reactor refuelling systems are dependent upon and are closely coupled with the nature of the fuel element to be used, the reprocessing and recycling concept adopted, the containment concept for the reactor plant, the reactor design, and the plant arrangement. The refuelling system and the reprocessing and recycling concept can have significant effects on the economics and plant availability of LMFBR plants. Extensive R&D effort will be required to develop and proof-test a satisfactory refuelling system, and in particular the fuel handling machines associated with this system.

6. Physics

Another area of fast breeder technology which underwent an intensive review in 1965 was physics. The results confirmed that significant gaps exist in our knowledge and understanding of the physics of fast breeder reactors.

Also, the review indicated the need for initial efforts to be aimed at acquiring data with sufficient accuracy for criticality and safety studies, and then gradually shifting to the gathering of more accurate data required for economic analysis.

The physics effort required to support the fast breeder programme is divided into three major areas, basic nuclear data, integral and critical experiments, and theoretical and analytical studies.

It is essential to have available basic neutron cross-section data over the entire energy range of importance for all materials used in fast breeders. There is a lack of available data because of past emphasis on thermal reactor technology, the difficulties of making measurements, and the magnitude of the job. The most serious gaps appear to be in capture and fission cross-sections in the range from 1·0 keV to 1·5 MeV, and inelastic scattering data above 1·5 MeV. These cross-sections directly affect the critical mass, flux energy and spatial distributions, breeding ratios, Doppler effects, and sodium void coefficients of fast reactors.

Measurements of the required cross-sections cannot be obtained in a reasonable time using existing facilities because a single cross-section measurement may take from a few months to several years, depending upon the neutron energy range to be covered and the precision desired. Therefore, it was considered essential to construct additional facilities to ensure that the necessary data will be available in time to provide adequately for the detailed design of the large fast breeder.

These additional facilities included the Oak Ridge Linear Electron Accelerator, and the Fast Neutron Generator at Argonne National Laboratory. Both are scheduled to be operable in 1968. Pending the availability of needed cross-section measurements, interim efforts will be required for theoretical prediction of cross-sections, refined evaluation of existing data, and estimation of cross-sections from the results of integral measurements.

At the present time, LMFBR's cannot be adequately designed using only basic cross-sections and existing computer codes; a large amount of data obtained from integral and critical experiments must also be used. The data obtained from these experiments relate to critical mass, danger or worth coefficients, Doppler coefficients, sodium void effects, neutron spectrum measurements, and dynamic behaviour. These experiments involve clean criticals, mockup criticals, and burnup experiments.

Clean criticals are generally small in size, contain a minimum amount of fuel, and are composed of a limited number of materials. The data obtained are used primarily to verify the calculational methods and cross-section data. Many such measurements have been or are being made. However, the interpretation of the results is hampered by the lack of knowledge of the neutron spectrum. Additional efforts are required to exploit spectrum measuring techniques such as time-of-flight and proton recoil.

As the basic physics becomes better understood, the experimental programme will shift from clean criticals to mockup criticals. These criticals are larger in size and geometrically more complex, and serve to verify the extrapolation from small to large systems as well as to provide direct information needed to design a demonstration plant. The major emphasis will be on understanding the temperature coefficients, danger coefficients, and dynamic characteristics of realistic reactors.

Theoretical methods required to understand LMFBR's are significantly different from those developed for thermal reactors. In particular, a more sophisticated treatment of the energy spectrum and transport effects is required. In addition, Doppler coefficients, heterogeneity effects, and multi-dimensional static and kinetic effects assume increased importance for LMFBR's.

Adequate calculation of the neutron energy spectrum requires the development of methods for processing basic cross-section data into multi-group sets, and the subsequent verification of these sets by comparison with data obtained from critical experiments. In addition, transport theory methods need further exploration, particularly questions concerning requirements for transport theory calculations and cross-section data. These studies will generally be confined to simple geometries and comparisons with clean critical results.

Increased attention will be given to the problem of predicting the dynamic behaviour of an LMFBR. Accurate prediction of the Doppler and other temperature coefficients as well as void coefficients are required. A large LMFBR is spatially weakly coupled and has a small feedback mechanism. Therefore, better multi-dimensional codes on core kinetics methods must be developed to predict the dynamic behaviour of cores in the FFTF, demonstration plants, and subsequent plants.

The development of theoretical methods will include multi-dimensional static and burnup methods to predict accurately isotopic changes and the reactivity at each position in the reactor and as a function of time. These methods are required for the economic analysis of the reactor as well as reactivity and safety analysis throughout the lifetime of the reactor.

Further details of the United States fast breeder reactor experimental and theoretical physics programme are presented in two other papers at this meeting.

Except for the availability of the FFTF and first demonstration plants, the major facilities required to meet the physics programme needs are expected to be in operation by 1968. It is expected that the test programmes of the demonstration plants will provide confirmation, on a conservative basis, of the various selected core configurations, and of the core physics and associated safety aspects. Increased use of foreign programme data is being obtained in part by the placement of representatives of the United States at Karlsruhe and Cadarache, visits of our leading engineers and scientists to other countries, and interchange of codes.

7. Safety

The Commission places primary emphasis on nuclear safety, and further intensive reviews are to be made of the LMFBR safety effort to identify and resolve the numerous safety questions and unknowns associated with fast breeder reactors.

The general safety criteria for sodium cooled fast breeder reactors recognize four significant factors: (1) the short neutron lifetime; (2) the use of plutonium with its resultant short delayed neutron reaction; (3) the hypothetically possible critical reassembly after meltdown; and (4) the use of sodium which introduces the problem of the sodium void coefficient and the possibility of severe chemical reactions with hydrogenous substances. Considering these factors, the goal of the fast reactor safety programme is to develop the technology and data needed for the safe design, siting, construction, and operation of fast power reactors. A secondary goal is to provide a sound technical basis for the continued evaluation of the design and safety of fast reactor installations. To attain these goals, maximum advantage must be taken of the experience available from water plant safety reviews. These include efforts pertaining to the establishment and implementation of design, construction, test and operations standards and practices, efforts to assure reduction in the probability of occurrence of misoperations or casualties which could finally result in the accidental release of fission products to the environment, and efforts directed at the development of engineered safeguards which permit the control of the consequences of accidents and the limitation of them to acceptable levels. All of these lines of effort are being vigorously pursued.

Some of the requirements can be identified by examining the sequence of phenomena occurring during hypothetical serious accidents in large ceramic fuelled sodium cooled fast breeder reactors. The subsequent course of events could include coolant flow blockage, sodium boiling, channel voiding, melting of fuel, progressive failure of adjacent fuel, possible fuel reassembly, power excursions, pressure pulses, vessel rupture, sodium–air reactions, fission product release, missile damage, effects on containment, and component and systems damage.

Examination and evaluation of such chains of events are needed to delineate areas of uncertain technology and methods and engineered safeguards systems for preventing or mitigating part or all of the accident.

Methods are needed to reduce fuel clad stresses and fuel clad embrittlement, and to increase fuel clad strength, to increase reactor stability, to detect incipient sodium boiling, to develop better understanding of sodium hydrodynamics and heat transfer which, in turn, will provide data required to design fuel rods and their assemblies, and to obtain better understanding of fuel meltdown and reassembly. Further, methods are needed to design the reactor, its associated safety systems, and containment, including missile shields, to better withstand the consequences of incidents. These and other developments are needed to eliminate or mitigate accidents.

Reactor containment requires considerable assessment. Though much has been done in the thermal reactor field, a limited amount of this will be applicable to fast reactor containment which introduces many different considerations. Some of these are plutonium and its fission products, the consequences of sodium–air reactions and related effects, the possible advantage of total inert atmosphere normally in the reactor containment, and post-incident cleanup.

This delineation illustrates the broad scope of the uncertainties which must be resolved to provide a firm technological base for the safety of liquid metal fast reactors, and to indicate the extent of the extrapolation of present small test reactor designs to the large sized reactors of interest to the nuclear industry. It is apparent that the design and construction of the first demonstration commercial plants will have to face and resolve the many safety problems to be encountered in designing a fast reactor.

Coupling is extensive between the safety programme and other LFMBR programme elements. For example, the major safety items in the physics programme include the determination of Doppler requirements, moderator requirements for degrading the neutron spectrum, fissile-to-fertile ratio for sodium void coefficients, and core configurations for sodium void coefficient as well as meltdown requirements.

Further, many of the items in the fuels and materials programme affect safety. These include determination of physical and chemical properties which may affect fuel and core safety, fuel behaviour, equations of state of the fuel, tests of fuel in TREAT type and possibly SPERT type facilities, and adequate fuel instrumentation to detect fuel failure and the course of transient events. Some of the major fuel properties which affect safety and require determination for all operating modes and abnormal plant conditions include thermal conductivity, expansion, enthalpy, compressibility, stoichiometry, and homogenization, In addition, determination must be made of these properties to satisfy the design of engineered safeguards systems.

In-pile data are needed to determine plutonium migration, void formation, slumping, clad-to-fuel compatibility, fission gas release, clad embrittlement, thermal cycling and ratcheting, swelling of restrained and unrestrained fuel, and the release of fission products, frothing, and fuel behaviour during and after meltdown.

In addition, the fast reactor must be protected against component and equipment malfunctions which may lead to serious operational losses and unsafe conditions. These include the malfunction of pumps and check valves resulting in loss of flow or overpressure; malfunction of control rods, fuel handling mechanisms, pump pressure transients and cold sodium accidents resulting in reactivity additions; piping, intermediate heat exchanger, or reactor vessel failures resulting in loss of sodium, steam generator tube failure resulting in a sodium–water reaction; and the loss of double containment which may result in a sodium–air reaction.

The results of the initial phase of the safety programme will be tested on a conservative basis to the maximum degree possible in the demonstration plants on an overall co-ordinated basis. Except to the limited extent that the FFTF will be able to contribute, the demonstration plants will provide the first instance where these results can be applied on the necessary scale.

8. Components

The immediate objective of the sodium components programme is to obtain components which will perform in a safe and reliable fashion at a cost which will be economically feasible for the utility application. The development of components which can operate safely and reliably in sodium containing impurities at temperatures up to 645°C, with cover gases saturated with sodium vapour and also containing impurities, is a task to which major resources and a large amount of time have been devoted; based on experience to date, this task will require significantly greater effort in the near future. Therefore, sodium components is another area which is undergoing an intensive review to delineate more specifically the development areas, to establish priorities, and identify the required resources.

As the size of the components gets larger and temperature differentials greater, the problems increase manyfold. As an example, the problems associated with the design, fabrication, and operation of a 410 m^3/min pump at dynamic heads of 107 m of sodium and temperatures over 370°C are considerably different than those for existing 67·5 m^3/min pumps. These requirements impose very difficult tasks on the industrial designers and fabricators who will play key roles in component development.

There is a pressing need to determine the criteria by which the design, fabrication, installation, and operation of a component can be assessed. Available standards must be used as a common area of understanding which can underlie the criteria; new standards must be established where needed. Specifications and drawings provide the component designer, materials supplier, fabricator, test technician, inspector, erector, and operator a co-ordinated engineering plan which defines all phases, from conception through operation, needed to ensure the requisite performance and a safe and reliable component. The development of instrumentation and inspection techniques which can non-destructively locate defects and potentially dangerous flaws must continue and be strengthened. Inspection of materials and components by qualified personnel who have been specially trained for the particular type of inspection must be a continuing affair. The absolute need for design confirmation and proof testing prior to installation in a costly plant cannot be overemphasized. It is the absence rather than the excess of such testing which is prevalent, resulting in setbacks to both the technology and the reactor plants. Operational testing in the plant cannot alone do the job; there is ample experience to prove that the use of reactor plants, rather than test loops, for verification of design and proof testing of components and equipment results in excessive costs and schedule delays.

In addition, the implications of such uncertainties and failures on the safety of the plant cannot be underestimated.

Proof testing invariably increases the possibility of timely identification of major weaknesses in components, and is an essential part of not only the LMFBR programme but all other Commission reactor development programmes. Major components such as the pumps, steam generators, intermediate heat exchangers, fuel handling machines, valves, and vital instrumentation should be part of a comprehensive proof test programme.

To provide a focal point for much of the proof and model testing, the Commission is negotiating a contract for a Sodium Component Development Centre at Santa Susana, California, operated for the Commission by Atomics International. Some of the facilities in this centre include the Sodium Components Test Installation (SCTI), the Control Rod Test Tower (CRTT), Sodium Component Test Loop (SCTL), and the recently authorized Sodium Pump Test Facility (SPTF). Commercial vendor utilization of this facility will become increasingly more important as the development programme progresses. Only a selected few of the most promising components can be installed in the first fast breeder reactor demonstration plants, and only limited development efforts can be conducted in these commercial power installations.

Further details in the U.S. Sodium Components Development Programme are being presented in another paper at this meeting. However, as was stated earlier, it is undergoing major review, and major adjustments will have to wait for the results.

It is important to note that sufficient detailed attention will also have to be given to component and systems engineering of relatively conventional-type devices whose failure may negate the value of costly instruments and equipment which have been developed under R&D programmes for the more exotic components.

9. Instrumentation and Control

The liquid metal instrumentation and control development programme is now under review as one of the major elements in the LMFBR programme. Its objectives are aimed at achieving successful instrument operation in, and control of, the fast neutron flux and liquid metal environment of the LMFBR test facilities and power plants.

Instrumentation development can be divided into three categories: (1) power plant information and control instrumentation; (2) power plant research and development instrumentation; and (3) special test instrumentation. Instrumentation in the first two categories must be designed and manufactured to high performance and reliability standards required adequately and safely to control a reactor plant and to obtain reliable and accurate information from it. The third category covers instrumentation used for highly specialized and usually short-term tests and laboratory experiments. It is recognized that considerable additional instrumentation will be installed, particularly in the first demonstration plants, to obtain technical information needed for the proper analysis and evaluation of component, equipment, and plant performance, for confirmation of design predictions, and for malfunction determination and analysis. The paucity of such instrumentation is evident.

The large and higher performance cores proposed for the LMFBR programme will impose a substantial increase in the amount of in-core data required, such as neutron flux, liquid metal flow, liquid metal temperature, fuel temperatures, and clad temperatures, increasing the problem complexity. The LMFBR programme test facilities will impose unique instrumentation and control requirements, such as fast response in-core temperature measurements, neutron spectrum analysis at power conditions, and liquid metal flow measurements at very high temperatures.

Although there have been no significant stability or safety problems with the operation of EBR-II, the larger cores proposed for future LMFBR projects will have different reactivity feedback effects because of a larger core size and probably a different core geometry. Temperature reactivity feedback effects include the Doppler, sodium void, and fuel expansion coefficients.

The sodium void coefficient is negative in small fast reactors, such as EBR-II, but may be positive, at least in portions of the core, for the larger fast liquid metal cooled reactors. The effect of the sodium void coefficient is significantly reduced by the apparent strong negative Doppler coefficient that should be characteristic of large fast reactor cores. This negative coefficient is prompt and does not depend upon fuel integrity and, thus, could contribute significantly to the safety of large fast reactors. Fuel expansion coefficients for a ceramic fuel are difficult to predict at this time.

At present it is considered impossible to predict accurately the magnitude and interrelationships of these reactivity effects in any new core design. Therefore, the results of the sizeable analytical and experimental effort being conducted in the physics and safety programmes to assess completely these parameters in large liquid metal cooled fast reactors must be factored into the design of control and safety instrumentation and systems.

Much of the instrumentation and control equipment required for the secondary sodium systems and turbine generation plant instrumentation and control equipment of large liquid metal fast breeder reactors has been developed. It can be produced by industry and, with an adequate quality assurance programme, should be expected to perform reliably.

Much of the instrumentation and control equipment associated with the reactor and the primary coolant system will require development for use in the anticipated environment of fast neutron flux of 10^{16} nv, gamma radiation of 10^9 R/h and liquid metal temperatures as high as 760°C. Detailed material, equipment, and process specifications have to be developed for industry to produce instruments with sufficient reliability and quality assurance.

A vigorous proof testing phase is required as part of the programme leading to the development of the equipment and instruments and their industrial standards and specifications; proof testing and disciplined quality assurance programmes are required for all equipment designed to operate in the LMFBR environment. Such a test programme will have to be established in time to assure significant benefit to the FFTF and to the first demonstration plants. As is the case with components, these initial plants will be using only a part of those instruments and equipment under development in the LMFBR programme.

Many types of special instrumentation will be required. For example, measurement of the internal temperatures of ceramic fuel elements is a problem yet to be solved satisfactorily. The temperatures involved can reach as high as 2800°C, and this precludes most of the industrially available thermocouple combinations. The tungsten–rhenium combination shows perhaps more promise for this application than other contenders, but has the disadvantage of undergoing a transmutation when exposed to neutron radiation which affects the thermoelectric properties of the elements. This change makes calibration of the thermocouple difficult. Development is currently under way to reduce the transmutation effects in addition to the effort on other promising thermocouple combinations that may be easier than the tungsten–rhenium combination. Although sodium temperatures are generally measured by the use of thermocouples, accurate but slow response resistance type thermometers have been used in some cases. The development of these instruments to operate to over 650°C is being undertaken.

Instrumentation for in-core measurement of fast neutron flux requires development. State-of-the-art neutron detectors are too large and have too limited a lifetime in the in-core environments proposed for the LMFBR programme.

Measurement of liquid metal flow has been accomplished with the flow tube and turbine type flowmeters as well as with the electromagnetic (EM) flowmeter. EM types have been more successful in the past, but troublesome with regard to their long-term accuracy, primarily because of the loss of magnetic field strength of the permanent magnets used in their construction at elevated temperatures. Today's magnetic materials are not adequate for operation above 650°C. Therefore, unless a better material is found, some other method of flow measurement in the high temperature test loops (to about 760°C) will be required. Investigations have been under way on turbine flowmeters, EM flowmeters, and on new principles of flow measurement, and such efforts will be augmented following completion of the review currently in progress.

Sodium level measurement devices include resistance probes, electromagnetic probes, ultrasonic detectors, gamma ray scanners, differential pressure sensors, and temperature sensors. Sodium contamination has been the chief cause of the problems encountered, so current development includes work on level probes that are more compatible with the liquid metal environment, as well as with techniques whereby no physical contact is made with the liquid metal.

Liquid metal pressure has been measured in a variety of ways. Existing pressure transducers operate reasonably well at temperatures below 540°C; development of these instruments is necessary to extend their operation to over 650°C.

Satisfactory operation of a liquid metal reactor at high temperatures for extended periods requires that the liquid metal purity be maintained at a very high level. It will be necessary to measure the oxygen, hydrogen, carbon, and possibly the nitrogen content of sodium with accuracies not previously attained commercially. Such instrumentation is beyond the state-of-the-art today. Work is under way to find new analytical techniques that would be easier to implement on an on-line basis than the methods currently available.

10. Sodium Technology

The sodium technology programme is also undergoing major review. Sodium technology support is necessary to assure that the fuel, components, instrumentation, and coolant systems are not adversely affected by the high temperature sodium and its impurities; the heat transfer and nuclear properties of the coolant remain within prescribed limits; any abnormal effects such as sodium–water and sodium–air reactions can be contained; the condition of the coolant and its cover gas can be measured and maintained within prescribed limits; the physical and chemical properties and the behaviour of the sodium, of the sodium impurities including fission products, and of the cover gas are determined; and that the methods of sodium and cover gas analysis and purifications can be determined to the necessary degree.

Accurate and reproducible methods must be developed for detecting less than 10 ppm of oxygen in sodium and for detecting and measuring many other sodium contaminants. The interaction between the cover gas and its impurities and the sodium and its impurities must be determined. New devices must be perfected to remove impurities to below acceptable levels.

Since a leak-proof steam generator has not been devised to date, it will be necessary to understand sodium–water–steam reactions to include the method of failure propagation and the subsequent course of events. Mass transfer, particularly from the fuel elements, requires a greater degree of understanding than accomplished to date, and the development work in this area has already increased.

11. Irradiation Facilities

As referred to previously, the extensive review, completed by the Commission in 1965, of the LMFBR fuels and materials development programme included the facilities associated with the development. A major finding was that there is a significant lack of realistic irradiation environment test space. In recognition of this, the irradiation facilities programme includes highest priority efforts on a Fast Flux Test Facility (FFTF), reconsideration of the use of the Sodium Reactor Experiment, organization of the FFTF efforts under the AEC's Pacific Northwest Laboratory as the AEC engineering laboratory project organization, orientation of the EBR-II to carry out fast flux testing, use of the Fermi reactor, utilization of the fast flux component of selected thermal test reactors, concentration of fast flux facilities, and increased use of data from foreign facilities.

Fast reactor fuels and materials, when irradiated in a thermal flux test reactor, will undergo effects which are different from those experienced in a fast flux test reactor. In a thermal neutron environment, most of the neutrons are absorbed at the periphery of the fuel element resulting in a non-uniform burnup: the result is a different fission product distribution, fuel depletion, temperature profile, and mechanical stress; the self-shielding effect practically precludes the testing of pin clusters or subassemblies in a thermal flux reactor; atom displacements and nuclear transmutations are different in the two environments (for example, helium and hydrogen production from fast neutron interactions with nickel and cobalt); and neutron dose and dose rates differ markedly between thermal and fast reactors. (For example, to obtain 10^{23} nvt greater than 1 MeV requires 5·5 years in the engineering test reactor and only 1·6 years in the proposed fast flux test facility, assuming a 60% plant factor.) The representative peak integrated flux in several U.S. reactors is shown in Fig. 3.

The previous lack of fast flux data, the resultant lack of correlation between available instrumental thermal flux data and the data to be available in the near future from relatively non-instrumented fast flux facilities such as EBR-II, the difficulties which will probably be encountered in attempting correlations as suggested above, and the need for realistic test data, require that an instrumented fast flux test facility containing closed loops be provided to test fast fuel elements in an environment as representative as possible of the LMFBR conditions.

As indicated previously, plans are now being made for the possible construction of a fast flux test facility to begin operations in the early 1970's. The major objective of the proposed fast flux test facility is to provide an adequate controlled and instrumented environment, in a representative LMFBR fast flux, for testing instrumented fuel specimens, fuel rods, fuel subassemblies, and clad materials. The plant should have capabilities for

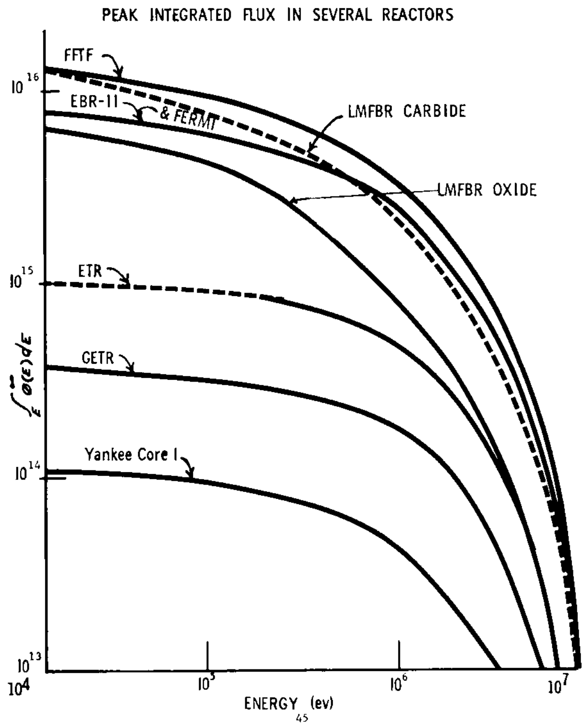

FIG. 3. Peak integrated flux in several reactors.

testing fuel up to and including failure in dynamic sodium; a reliable and safe plant performance (high and predictable plant factors); capabilities to test fuel for short time periods; and capabilities for non-destructive interim fuel examination. The facility will serve all U.S. fast flux requirements to the maximum extent possible.

To meet the objectives and provide the capabilities, the Commission now has under consideration a range of requirements which include four to eight instrumented closed loops, each up to 6 in. o.d.; facilities for short time irradiations; provisions to meet programmatic needs for planned programmes other than the LMFBR; facilities for interim fuel examination, including disassembly, reinstrumentation, and reassembly of fuel specimens, capsules or rods; a reactor flux ranging from 0·7 to $1{\cdot}3 \times 10^{16}$; closed loop sodium temperatures to 760°C; reactor core outlet to 650°C; and the means to measure and control sodium impurities. The neutron flux level of about 10^{16} requires a plant power level of 300–400 MW(th); and the size and number of closed loops together with flux levels requires a reactor core volume of 600–800 litres.

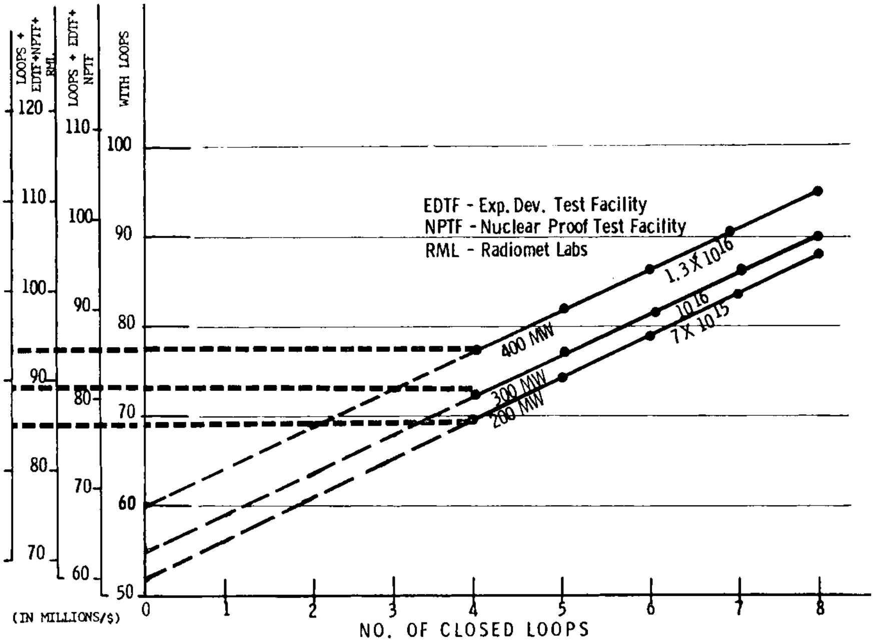

FIG. 4. Fast flux test facility (capital costs).

Other requirements include: a minimum of two reactor primary system loops, two secondary loops, and two heat sinks, designed to permit continued plant operation with one loop in a non-operating status; closed loops accessible and removable external to the reactor; adequate provision for closed loop coolant control instrumentation and monitoring equipment, including all equipment for controlling and/or monitoring the test specimen, capsule, rod or subassembly; adequate provision for in-core instrumentation and control equipment; accessibility from the reactor exterior to make and break instrumentation leads; and a goal of 75% of the plant availability factor.

The present plan now being implemented for the FFTF has as its immediate goal the definition of the reference design required to best meet the LMFBR objectives, including the FFTF's supporting facilities, R&D programmes, schedule, related resources (test facilities, costs, and personnel), and project management structure. The summer of 1966 is the scheduled completion date for this phase.

Parametric studies of the FFTF are now being conducted to include parameters such as flux, spectrum, type of driver fuel, type of control, reactor power, temperatures, reactor core temperature difference, number of closed loops, size of loops, type of reactor design, and the number of heat removal loops. The conditions affected will be evaluated to include test requirements, plant availability, safety, and costs. The design will be developed to obtain an optimum for the moneys allotted. Figure 4 is an example

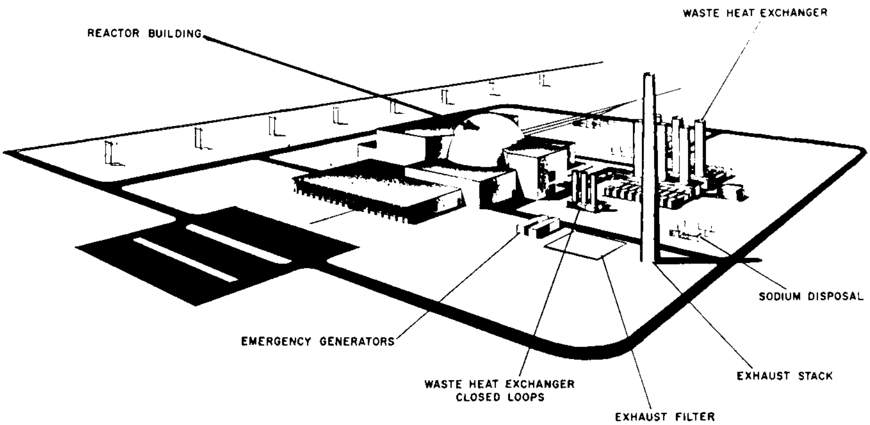

FIG. 5. Fast flux test facility.

of how parametric variations could affect the cost of a fast flux test facility.

The successful undertaking and timely accomplishment of a safe and reliable facility with a minimum complexity and cost which will meet the above requirements, including systems and components (e.g. fuel handling, closed loops, instrumentation, pumps, steam generators), can be assured only by a substantial and competent development and engineering effort from the AEC–industry–laboratory complex. Moreover, it will require a more thoroughly planned, disciplined, and extensive effort than has been characteristic in the past successful development of some important programme facilities that are similar but not as complex. It is well recognized that the facility will be difficult to undertake, that large size leads to large cost, and that certain complexitics cannot be avoided. This does not deter the Commission from accomplishing the task necessary to meet the programme objectives and provide the key facility to realize the important basic R&D objectives.

Figure 5 is the reference concept now being considered for the FFTF. Some of the major problems associated with the design of the FFTF are its closed loop design, closed loop accessibility for replacement, closed loop safety, driver fuel development, fuel handling design and development, instrumentation, and in-core instrumentation accessibility for disconnecting and reconnecting.

Closed loops, the key item in the facility, have major problems which include requirements that the in-core tubes be accessible and removable from outside the reactor, instrumentation connexions must be accessible from outside the reactor for disconnecting and reconnecting, and the design of the in-core tubes must be such as to accommodate fuel meltdown without physical interaction with adjacent areas. The design of the in-core tubes must be such as to assure no adverse or unsafe reactor conditions due to reactivity excursions which may occur in any of the experiments in the loops, recognizing that the experiment itself will have to be designed to provide the same assurance.

12. Use of Existing Irradiation Facilities

The prime purpose of EBR-II, the only AEC owned fast facility in operation, has been changed from demonstrating an integrated reactor–fuel recycle system to a high priority fuels and materials test facility in direct support of the LMFBR programme. A review of EBR-II is being made to determine those factors which may affect its irradiation test capabilities, including plant availability. Steps will be initiated to correct any deficiencies which are found. The evaluation will include operating and maintenance schedules, driver fuel criteria, safety of fuel tests, test priorities, effects of full power operation, and methods for extrapolating post-irradiation fuel examination data to an improved core operation.

The Fermi reactor, an industry-owned fast flux facility, is estimated to be available for irradiation testing by January 1967 on the basis that the power test programme will be completed during the summer of 1966. The Commission has reached agreement for the use of Fermi reactor test space. Consideration of such use has taken into account many factors, including driver fuel limitation, contractual limitations and plant availability.

Additional reviews are under way to determine the availability of other reactors in the United States and abroad, such as the Sodium Reactor Experiment (SRE), and to determine what part they can play in the AEC fuels and materials irradiation test programme. For example, the 30 MW(th) SRE had been redesigned previously as a part of the sodium–graphite programme to provide a high temperature sodium environment up to 650°C. The current review is being made to determine the capabilities of SRE to meet some of the immediate needs of the fuels and materials programme, such as screening of fuels. The low neutron flux of SRE limits the amount of irradiation data which can be acquired. The use of other thermal flux reactors not cooled with sodium will likely be limited to utilization of the fast flux component.

Table 2 lists the characteristics of some of the existing and planned U.S. facilities, including the FFTF.

TABLE 2. U.S. IRRADIATION FACILITIES DATA

Unit	EBR-II	Fermi	FFTF*	SRE-PEP
MW(th)	62·5	200	400	30
$Xnv \times 10^{-16}$(max)	0·36	0·5	1·35	0·01
kW/kg Pu	670	780	$\sim$2000	1000
Core:				
Litre	73	380	600	4800
Dia. cm	50·8	83·8	91·5	183
Height cm	35·5	83·8	86·5	183
°C core outlet	470	430	650	650
Driver fuel	U-5 Fs	U-10 Mo	PuO_2-SS	UC
No. reactor loops	1	3	$\geqslant$2	1
No. closed loops	0	0	6	0

* Tentative.

13. Use of Demonstration Plant as an Irradiation Facility

The design and construction of demonstration utility plants is a vital element of the LMFBR programme, and in this environment major demonstration and operational evaluations of the most promising products of the R&D programme will be accomplished at the time of their availability.

The use of this plant as a test facility for the conduct of the comprehensive fuels and materials development and testing programme planned would require extensive and difficult provisions for instrumented closed loops. In this event, the design requirements for such testing could seriously compromise the intended purposes of the demonstration plant. The LMFBR programme is sufficiently important that the successful demonstration of a power plant should not be jeopardized or compromised for this or similar basic technological developmental purposes.

In addition to the design restrictions imposed on the fuels, structural materials, and the power plant by the severe safety considerations related to LMFBR plants, the first demonstration plants will have to be built to a conservative design and operated at conservative conditions, based on results of prior development efforts. Consequently, the first demonstration plants may be designed to accommodate some fuel difficulties, but they will not be able to afford an extensive core failure, even though safe, regardless of the cause of failure.

14. Plant Design

The results of the 1000 MW(e) studies performed by industrial firms within the past 2 years are being factored into follow-on studies initiated by the Commission. These studies will include parametric analyses to examine the relationships among the many design and operating parameters with a view toward selecting the most promising concepts for further study and development. These studies will identify needs that are common to all alternate design concepts, or nearly so, and will in this way identify any additional priorities on requirements for research and development work to achieve programme objectives. Many of these requirements have been identified early in the study programme and development work need not await the final results of the studies. The studies will weigh the various tradeoffs and will determine the relative advantages of the various technical approaches in terms of performance and cost with respect to achieving the various development objectives. These results should assist in making determinations regarding the application of national resources (people, facilities, and funds) to those areas most likely to result in the greatest overall benefit to the programme. These studies will also be used to define better the role that demonstration plants can play in the R&D programme. In addition these studies will help assure development of a system and plant design and engineering capability in a number of industrial contractor organizations. Meaningful inputs from experienced architect-engineers and utilities will be required as these studies progress. The studies will result in a reference design concept for a 1000 MW(e) plant and the results of the reference design will be factored into the design of the first demonstration plants.

15. LMFBR Demonstration Plants

The history of the civilian power water cooled reactor programme has shown that in addition to programme oriented test facilities, a number of demonstration plants are essential to validate the results of the overall R&D programme, to provide an industrial base, and to obtain acceptance of the plant type by the utilities. Similarly, demonstration plants will be used for the LMFBR programme.

The first demonstration utility plants included in the overall LMFBR plan will have a capacity somewhere in the 300–500 MW(e) range, will be an integral part of a utility power grid, and will prototype and demonstrate as many features as possible of a 1000 MW(e) or larger plant.

This size plant and the related investment are significantly larger initial steps than those taken previously in the commercial nuclear power demonstration programme in the U.S. In addition, there are many other considerations that alert us that this LMFBR programme will be technically difficult and expensive to achieve. These include the following:

(1) Although sodium cooled reactor technology is relatively advanced with respect to other fast reactor concepts, it requires significant augmentation to approach the state of the pressurized water cooled thermal reactor technology. For example, in the early days of commercial exploitation, water reactor programmes had a good base in areas such as components and water chemistry, and in the successful navy plants. Furthermore, the extensive effort expended in the development of the production reactors resulted in significant contributions to the thermal reactors programme (e.g. xenon poisoning discovered at Hanford).

(2) The use of sodium in experiments and engineering tests has required special procedures and precautions, leading to more stringent requirements in design, operation, and maintenance of components, systems and plants. Operation of EBR-II, Fermi, Hallam, and SRE has indicated the need to strengthen considerably these procedures and precautions and add new ones, particularly in view of the increases over conditions previously encountered in performance requirements, such as temperatures and radiation exposures.

(3) The operations at existing sodium cooled plants, the increase in proposed LMFBR plant size, and the increase in operating conditions, such as temperatures, flux, and fuel burnup, have indicated that the fuel and components for future plants will not be engineering extrapolations but truly the first of their kind. These will require extensive and expensive design confirmation and proof testing.

(4) The high performance and unique characteristics of LMFBR plants require the development of new engineering codes and standards in addition to the presently available base.

(5) The LMFBR will require large inventories of highly subdivided enriched plutonium–uranium fuel fabricated at costs currently estimated at at least three to five times per unit of heavy metal over that of the present fuel costs for commercially available water plants.

(6) LMFBR power densities are five to ten times those in commercial water reactors, flux levels one to two orders of magnitude higher, and fuel burnup two to five times greater.

(7) The bulk reactor outlet temperature will be 540°–650°C as compared to less than 315°C for water reactors.

(8) The fuel clad temperature range in the LMFBR will be 650°–760°C as compared to 315°–345°C for water reactors.

(9) The effect of a fast neutron spectrum and high temperature on fuels and other materials under high flux conditions in sodium over long irradiation periods introduces problems of complexity hitherto unknown.

(10) Plutonium bearing fuels will need to be developed and tested under LMFBR conditions such as those noted above.

(11) Fast reactor statics and transient physics characteristics must be obtained for large core loadings containing up to several thousand kilograms

of plutonium. These tests will be extremely expensive. Up to $21,000,000 is estimated for fabricating a critical experiment loading for the ZPPR.

(12) LMFBR cores and components must be developed to withstand high temperatures, thermal gradients and shocks unique to liquid metal cooled reactors.

(13) The complex interrelationships of fast reactor nuclear parameters, not as well understood as comparable relationships for thermal reactors, require that a broad programme in fast reactor safety be initiated.

The first demonstration plants will have to prototype and validate technical understanding and engineering control of most of these features. Many other important aspects of the LMFBR programme can only be confirmed in a demonstration plant in the utility environment. These include demonstration of overall fuel cycle, long-term reliability and safety of operation, and high plant factor and maintainability, along with related potential economics of a system designed for and operated by utilities. As stated previously, such plants represent the focal point of the immediate R&D programme, and these will be the ultimate tests of the initial phase of the overall programme.

In looking forward to the commitment for the first demonstration plants, the status of certain key programme elements must be considered. These include:

(a) Proof test of fuel, fuel clad, fuel support structure, and fuel subassembly to over 50,000 MWd/t and over 10^{23} nvt.
(b) Successful development and demonstration of key components.
(c) Resolution of key sodium technology problems.
(d) Successful operation of EBR-II or Fermi at adequate plant factors.
(e) Resolution of key physics problems requiring use of cross-section accelerators, critical assemblies, and SEFOR.
(f) Resolution of key safety problems requiring use of critical assemblies, TREAT, and SEFOR.
(g) Development of the industrial capability.
(h) Meaningful industrial and utility participation.

Key test facilities need to be in operation and test results available.

The progress of the R&D programme and the development of competitive industrial capabilities will help determine the time of the commitment to the first large LMFBR demonstration plants. The decisions on building such plants will not only depend upon realistic assessment of the technological and engineering uncertainties, but also on the associated economics, the financial risks, and the willingness of reactor manufacturers, utilities, and the AEC to participate in accepting the risks involved. Based on current evaluations of these many factors, it does not appear that a decision to commit to the first U.S. demonstration plant can be made before 1969.

16. Other Coolants for the Fast Breeder Reactor

There have been several studies in the U.S. proposing the use of coolants other than sodium for the fast breeder reactor, and follow-on studies are currently in progress.

Both the steam cooled fast breeder and the gas cooled fast breeder must be considered as candidates. Therefore, a minimal development effort is being devoted to these and other alternative breeder concepts. Emphasis will be placed on the fuels, physics, safety, and plant conceptual design work. This R&D work is considered to be in its early stages and the related prerequisite technology provides sufficient reasons for proceeding with deliberate actions before undertaking a full scale effort. The steam cooled concept would make maximum use of water plant technology and components and the familiarity of the utilities with water–steam plants, both conventional and nuclear. In addition, the possibility exists of employing a direct cycle such as is used in the boiling water reactor. The problem areas are in the fuel, physics, safety and those areas related to the use of a moderating coolant while attempting to retain good breeding capability. The gas cooled fast reactor concept would make use of technology which has been highly developed in the United Kingdom and the United States. The application of the U.S. High Temperature Gas Reactor (HTGR) work to a fast gas reactor involves substantial extrapolation, particularly in the areas of fuel and core design, performance, and safety. It appears unlikely that the economic and technical comparisons will give any significant economic or broad engineering advantages to either the steam or gas cooled concept to the degree necessary to provide reasons for changing the high priority assigned to the sodium cooled fast breeder reactor in the U.S.

17. Conclusion

In its programme to achieve systems that will extend the United States' energy resources while reducing further the cost of electrical power, the AEC has given the highest priority to the liquid metal cooled fast breeder reactor. The potential economic advantages far outweigh the difficult development problems and the expenditures currently proposed; the cost benefits of the extensive R&D programme are obvious. The successful demonstration of this vital reactor concept in a timely, effective, and economic manner depends upon verification of the technological and engineering prerequisites through a planned and disciplined research and development programme. The required elements to assure a successful R&D programme are summarized in Fig. 6. A successful R&D programme should culminate in LMFBR demonstration plants in the utility environment followed by larger economical sodium cooled fast breeder reactor power plants.

Significant and valuable quantities of R&D information have been made available to participants in this conference under the interchange arrangements of agreements for co-operation. However, conferences such as this

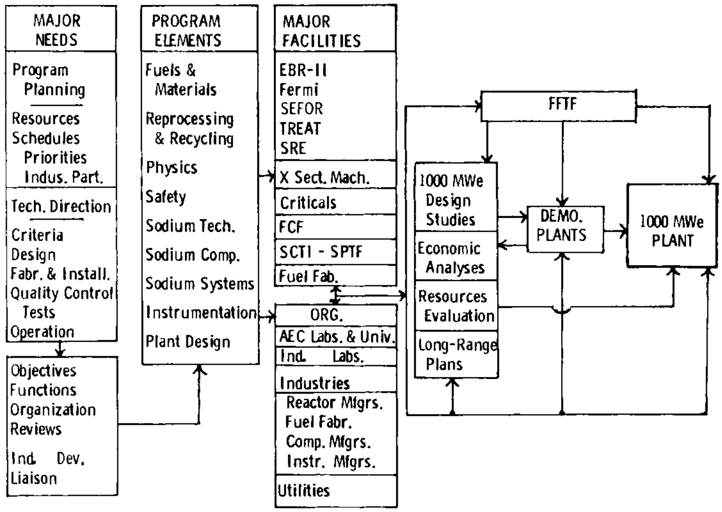

FIG. 6. The path to the LMFBR objective.

one personalize these exchanges and update the information. It is our hope that these technical forums continue to provide us with increased assurances based upon a better understanding of the LMFBR programme and the tremendous incentives related to its timely accomplishment.